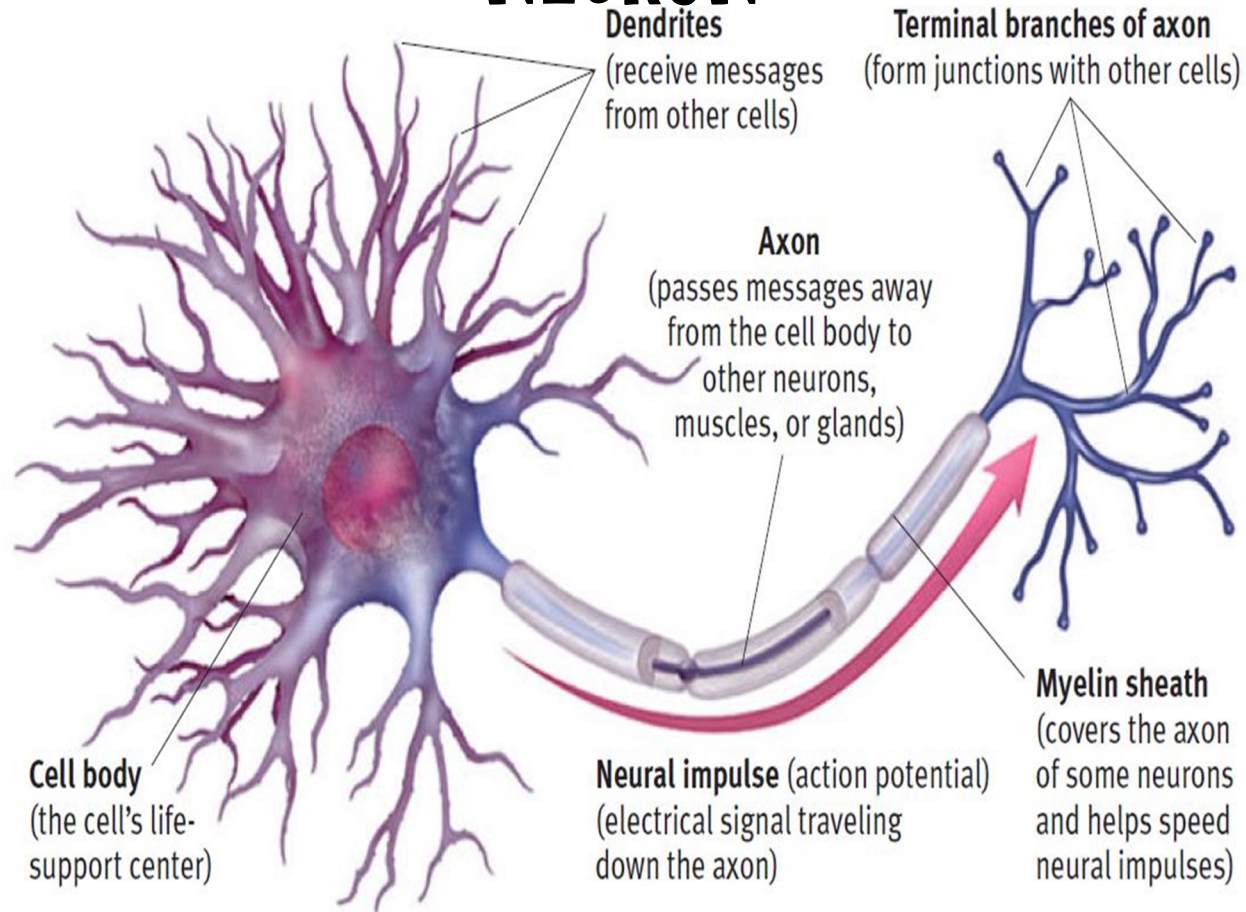


NEURON



What are the different types of neurons?

Sensory



These neurons take sensory information from the environment (ie. getting poked with a needle) and sends the signal to the brain.

Motor



These neurons communicate information from the brain to tissues and organs throughout the body, allowing for movement.

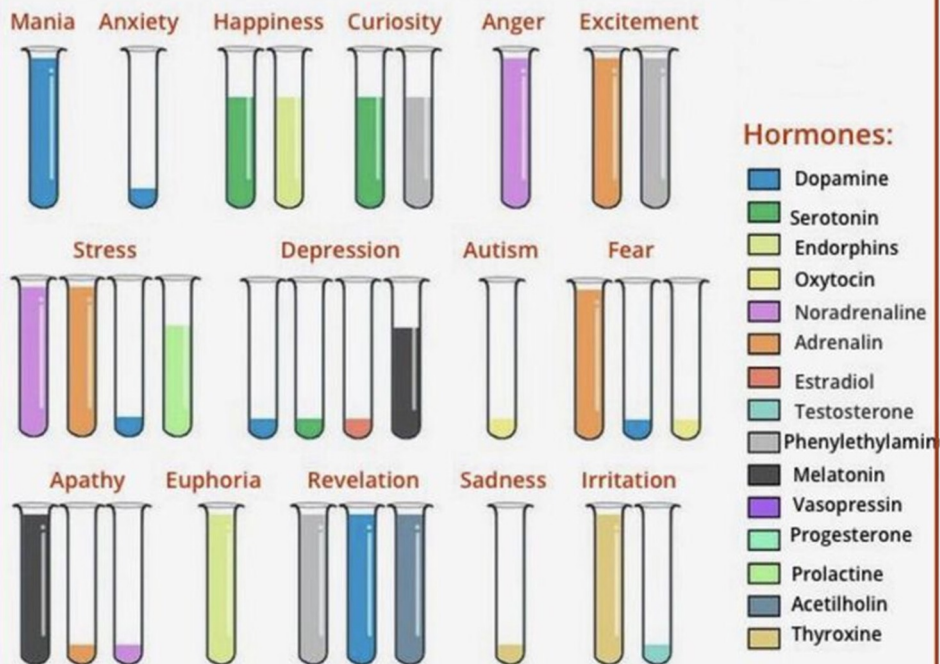
Interneuron



These neurons make up the majority of neurons in the body. They are essentially the "middle man," transmitting information between sensory and motor neurons. They play a key role in learning, memory, and planning!

Neurotransmitter	Function	Examples of Malfunctions
Acetylcholine (ACh)	Enables muscle action, learning, and memory	With Alzheimer's disease, ACh-producing neurons deteriorate.
Dopamine	Influences movement, learning, attention, and emotion	Excess dopamine receptor activity is linked to schizophrenia. Starved of dopamine, the brain produces the tremors and decreased mobility of Parkinson's disease.
Serotonin	Affects mood, hunger, sleep, and arousal	Undersupply is linked to depression. Prozac and some other antidepressant drugs raise serotonin levels.
Norepinephrine	Helps control alertness and arousal	Undersupply can depress mood.
Endorphins	Lessen pain and boost mood	If flooded with artificial opiates, the brain may stop producing endorphins, causing intense discomfort.

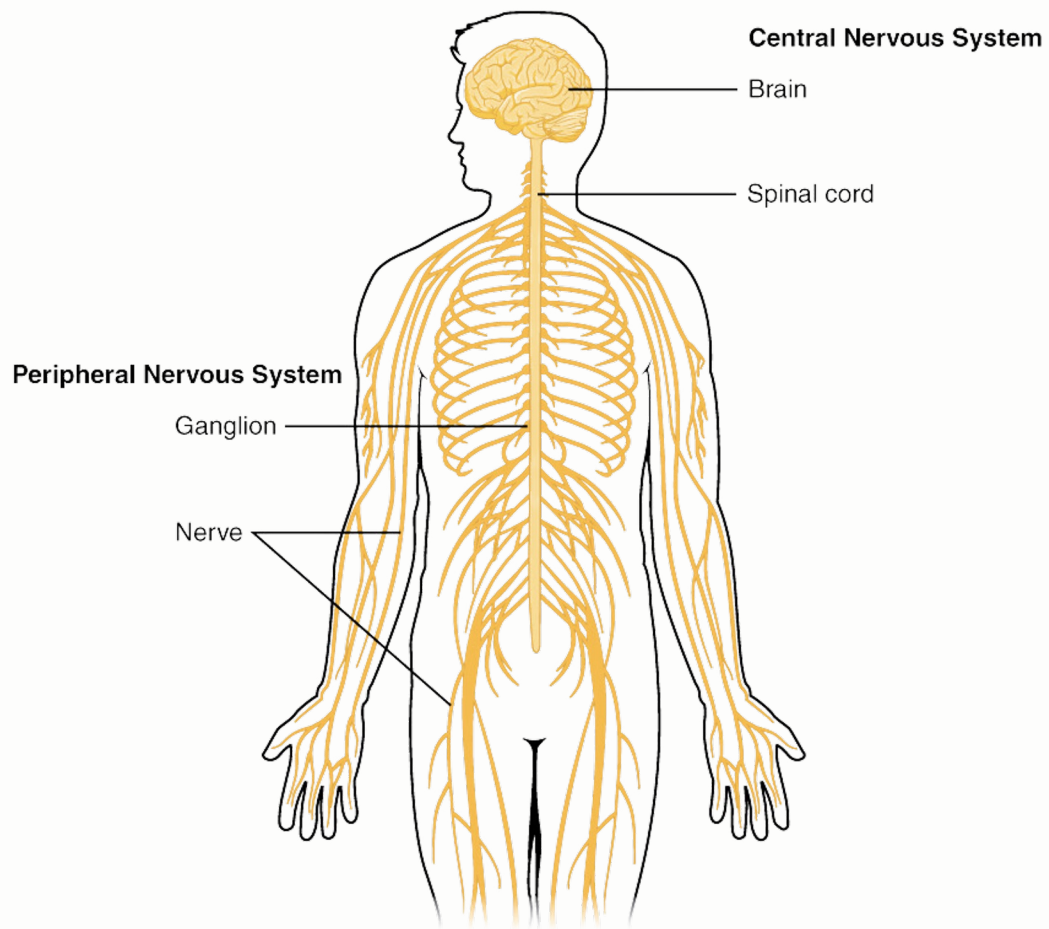
Emotions & Chemistry of Hormones



THE NERVOUS SYSTEM

The central command store of our body.

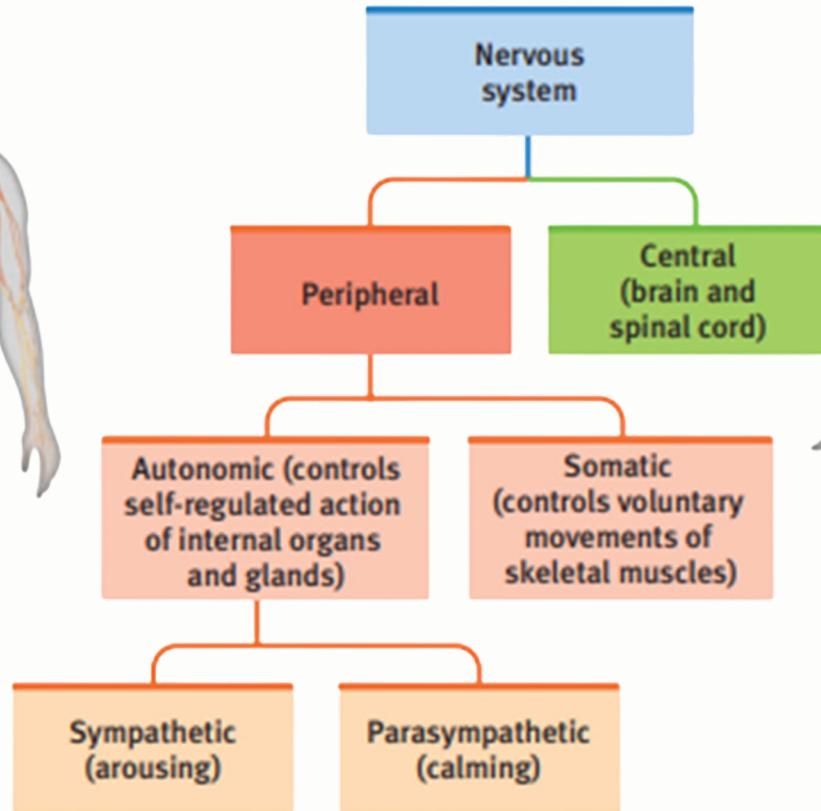
- Thoughts, memory, learning, and feelings.
- Movements, such as balance and coordination.
- Senses, including how your brain interprets what you see, hear, taste, touch and feel.
- Sleep, healing and aging.
- Heartbeat and breathing patterns.
- Response to stressful situations.
- Digestion, as well as how hungry and thirsty you feel.
- Body processes, such as puberty.



Peripheral nervous system



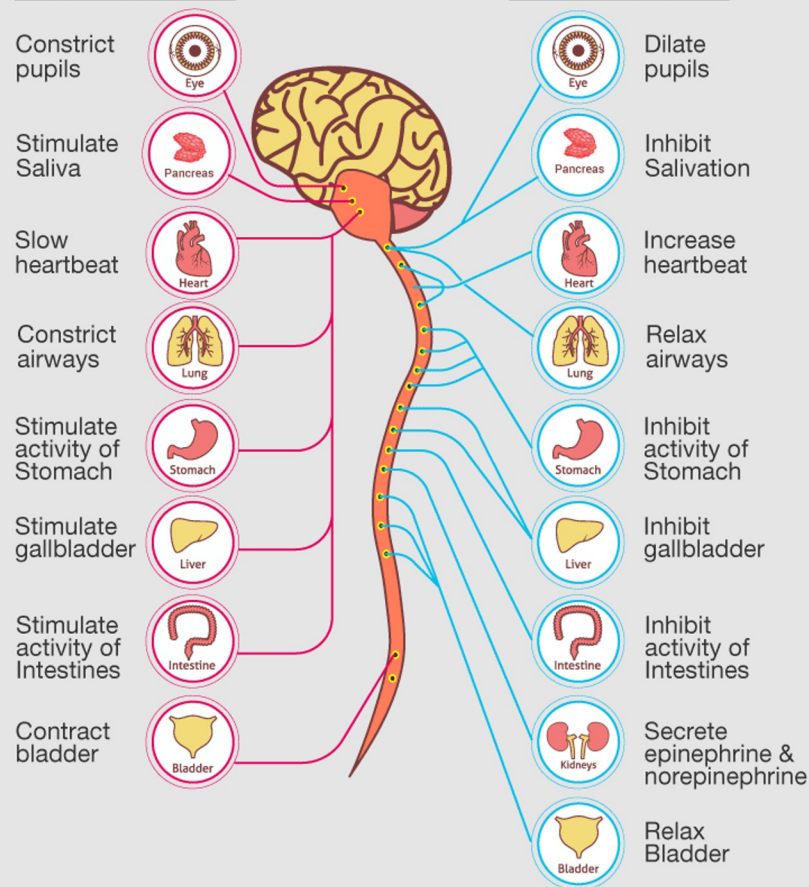
Central nervous system

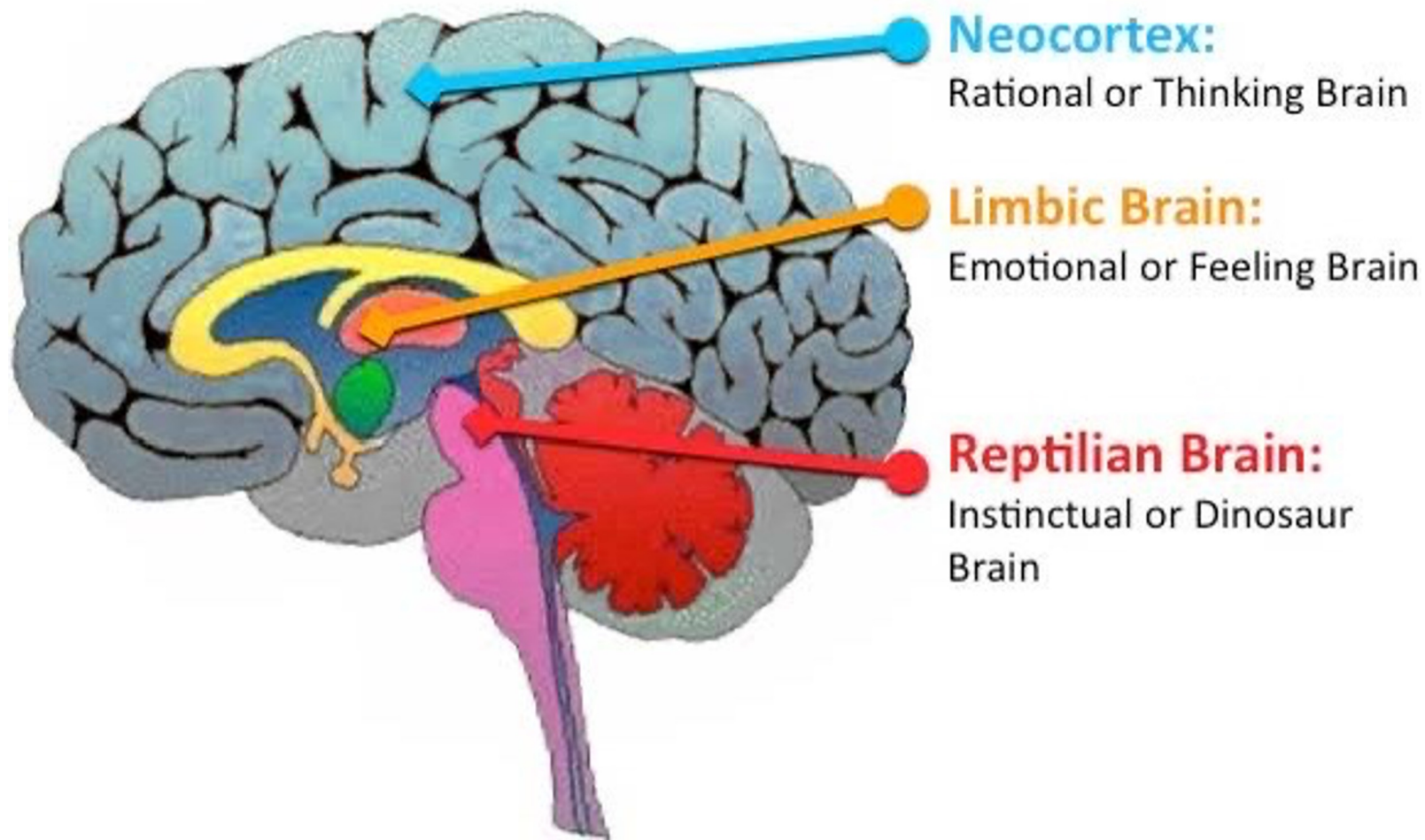


PARASYMPATHETIC NERVES

Vs

SYMPATHETIC NERVES



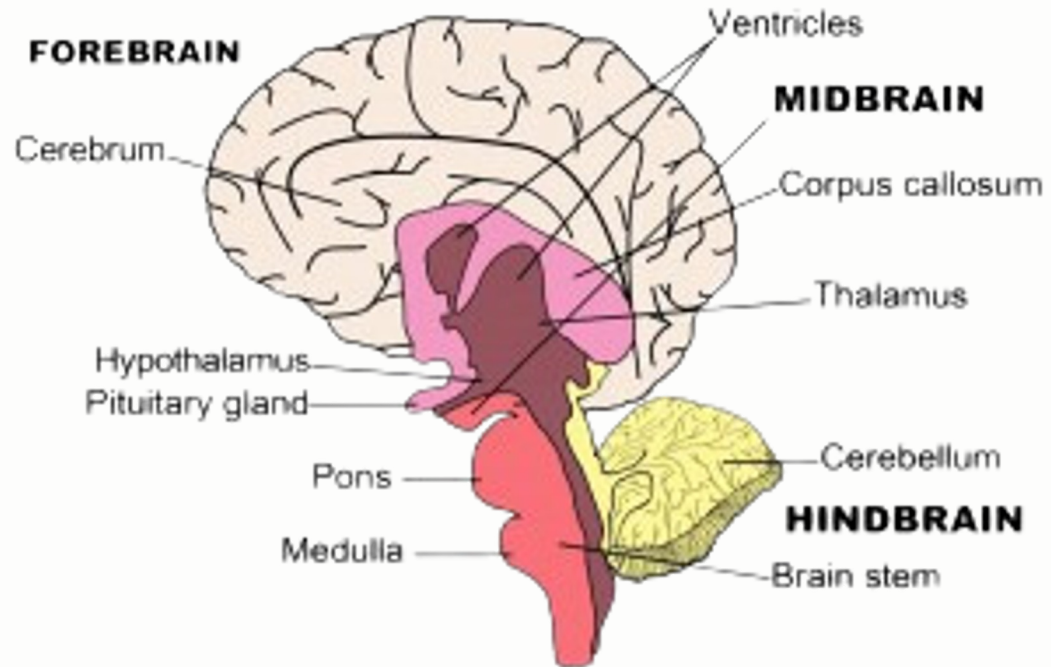


THE ORGANIZATION OF BRAIN

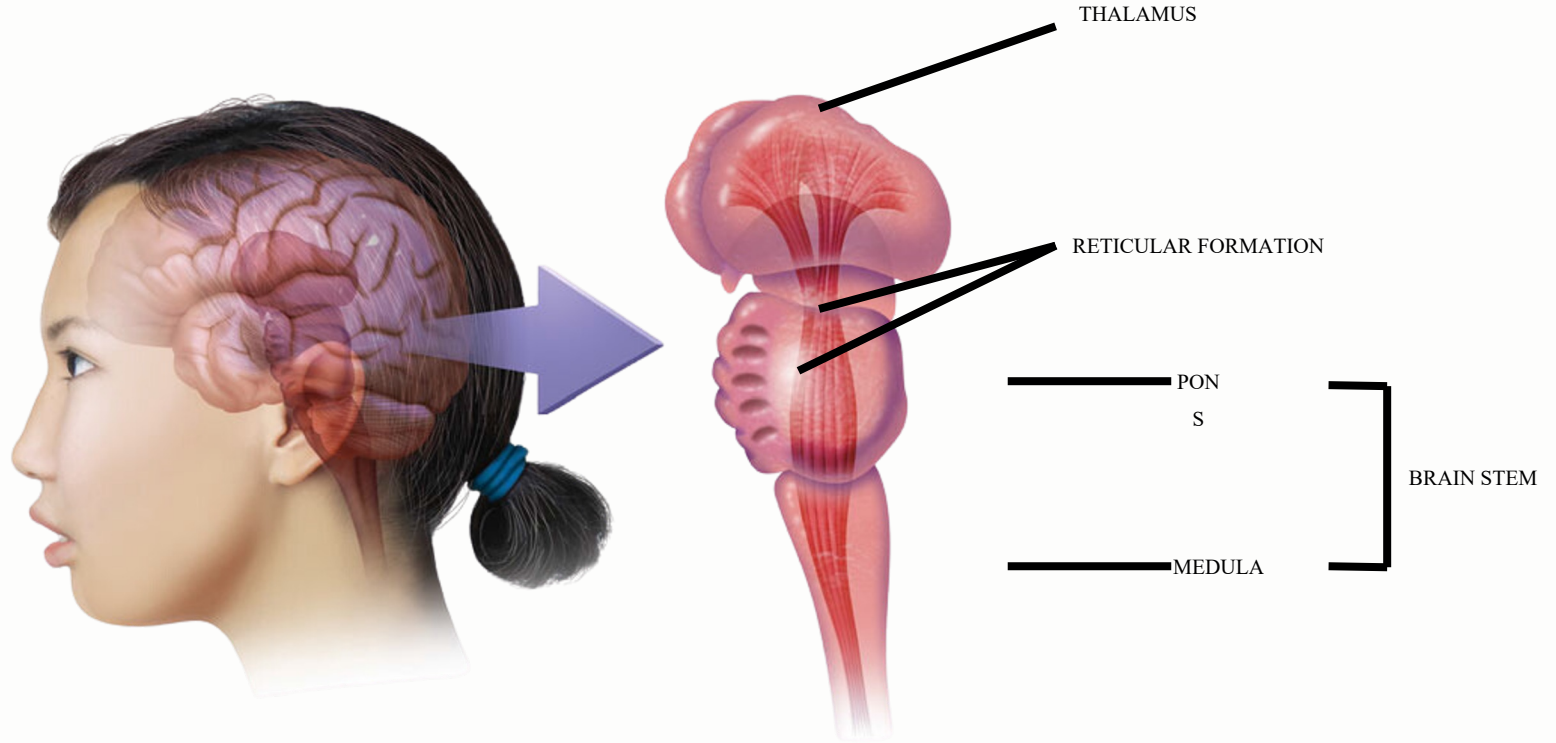
1. CEREBRAL CORTEX

2. LIMBIC SYSTEM

3. THE BRAINSTEM

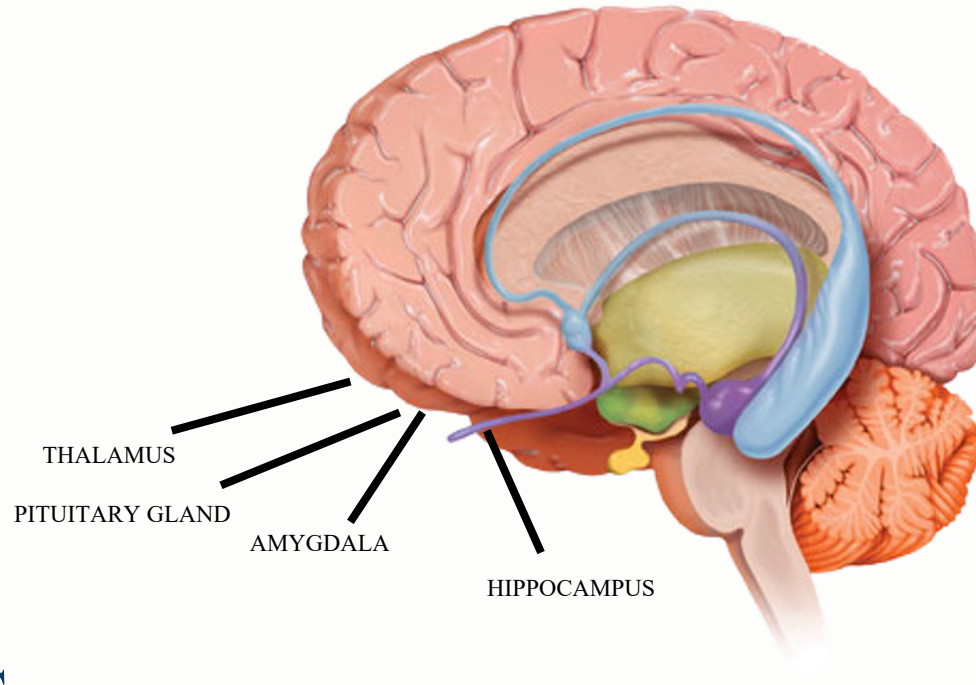


THE BRAIN STEM

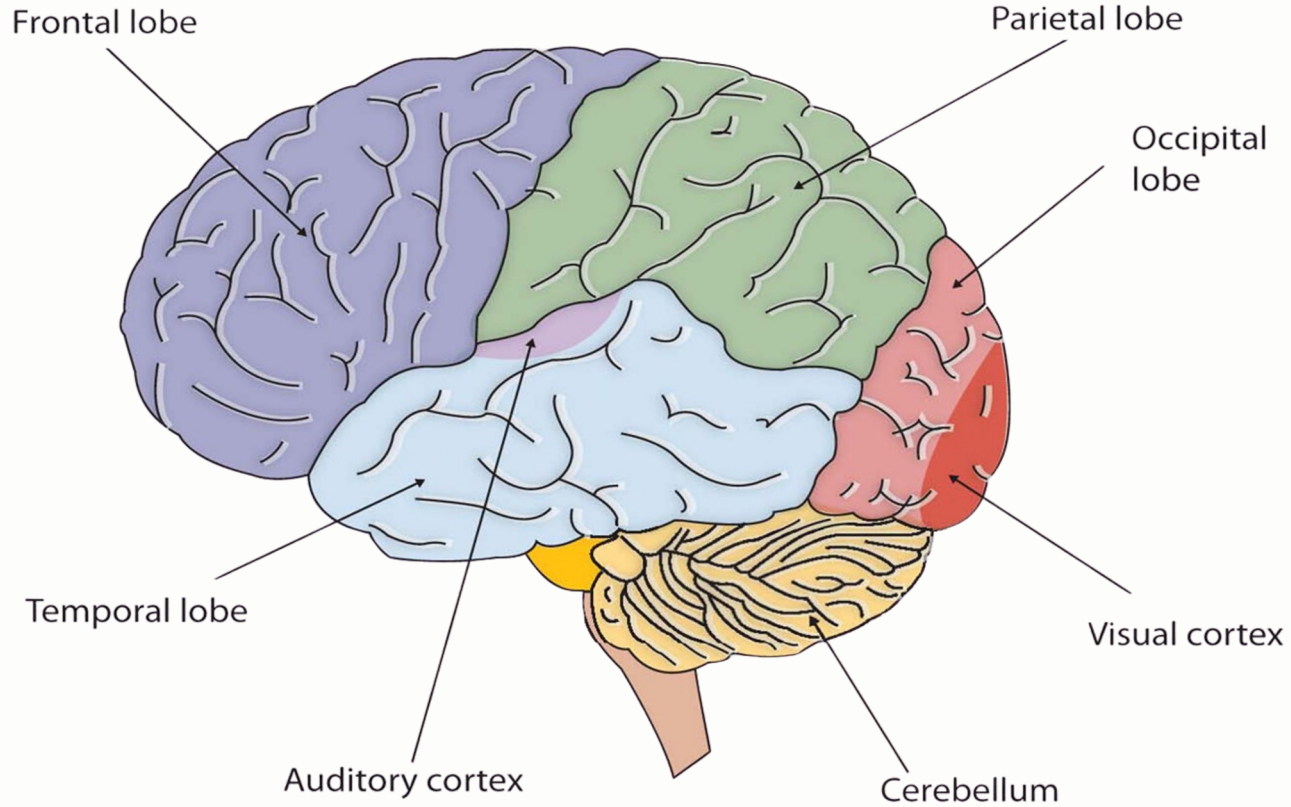


THE LIMBIC SYSTEM

1. THALAMUS
2. PITUITARY
GLAND
3. AMYGDALA
4. HIPPOCAMPUS

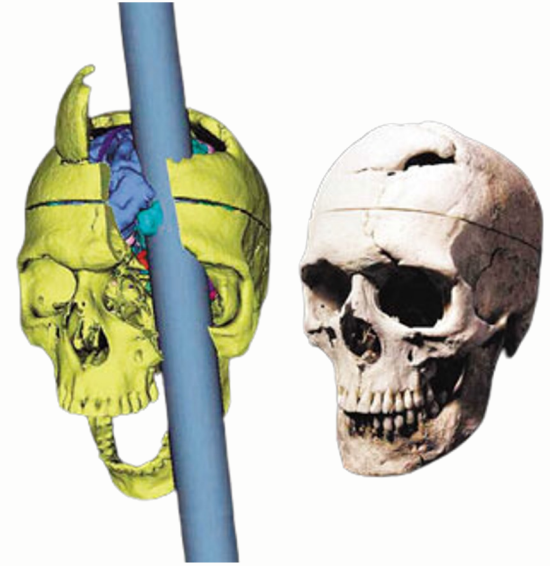


THE CORTEX

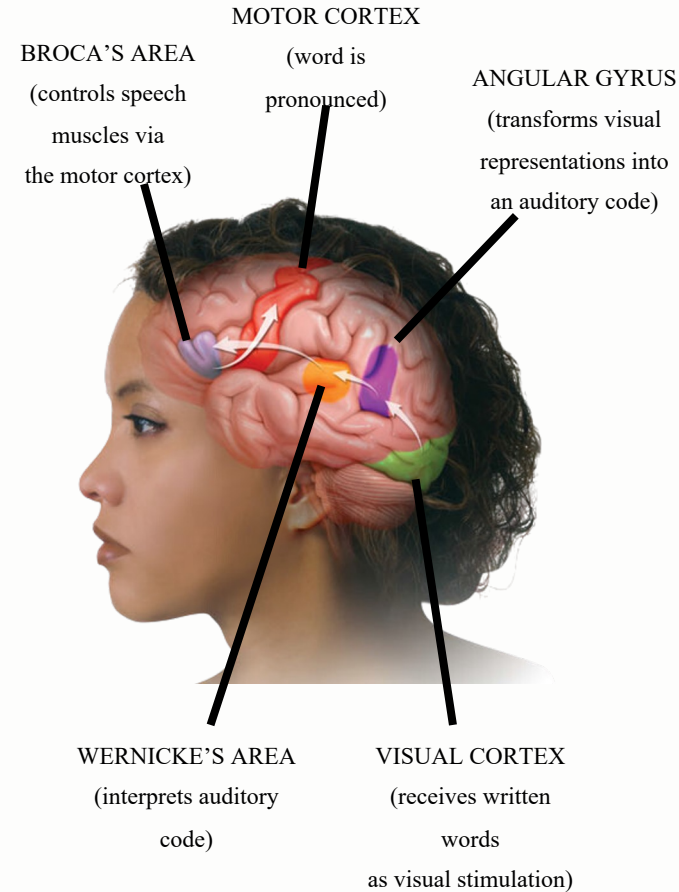


PHINEAS GAGE

One afternoon in 1848, Gage, then 25 years old, was packing gunpowder into a rock with a tamping iron. A spark ignited the gunpowder, shooting the rod up through his left cheek and out the top of his skull, leaving his frontal lobes massively damaged. To everyone's amazement, he was immediately able to sit up and speak, and after the wound healed he returned to work. But the affable, soft-spoken Phineas Gage was now irritable, profane, and dishonest. Although his mental abilities and memories were intact, his personality was not. This person, said his friends, was "no longer Gage." He eventually lost his job and ended up earning his living as a fairground exhibit.



- **Aphasia**, an impaired use of language, can result from damage to any one of several cortical areas.
- In 1865, French physician **Paul Broca** reported that after damage to a specific area of the left frontal lobe (later called **Broca's area**) a person would struggle to speak words while still being able to sing familiar songs and comprehend speech. Damage to Broca's area disrupts **speaking**.
- In 1874, German investigator **Carl Wernicke** discovered that after damage to a specific area of the left temporal lobe (**Wernicke's area**) people could speak only meaningless words. Asked to describe a picture that showed two boys stealing cookies behind a woman's back, one patient responded: "Mother is away her working her work to get her better, but when she's looking the two boys looking the other part. She's working another time" (Geschwind, 1979). Damage to Wernicke's area also disrupts **understanding**.
- A third brain area, the **angular gyrus**, is involved in **reading aloud**. It receives visual information from the visual area and records it into an auditory form, which Wernicke's area uses to derive its meaning. Damage to the angular gyrus leaves **a person able to speak and understand, but unable to read**.



1. How do you approach problem-solving?

- I rely on intuition
- I break the problem down into smaller steps and use logic.

2. How do you process information better?

- Through images, patterns, and stories.
- Through numbers, facts, and clear sequences.

3. When learning something new, do you prefer hands-on experience or reading detailed instructions?

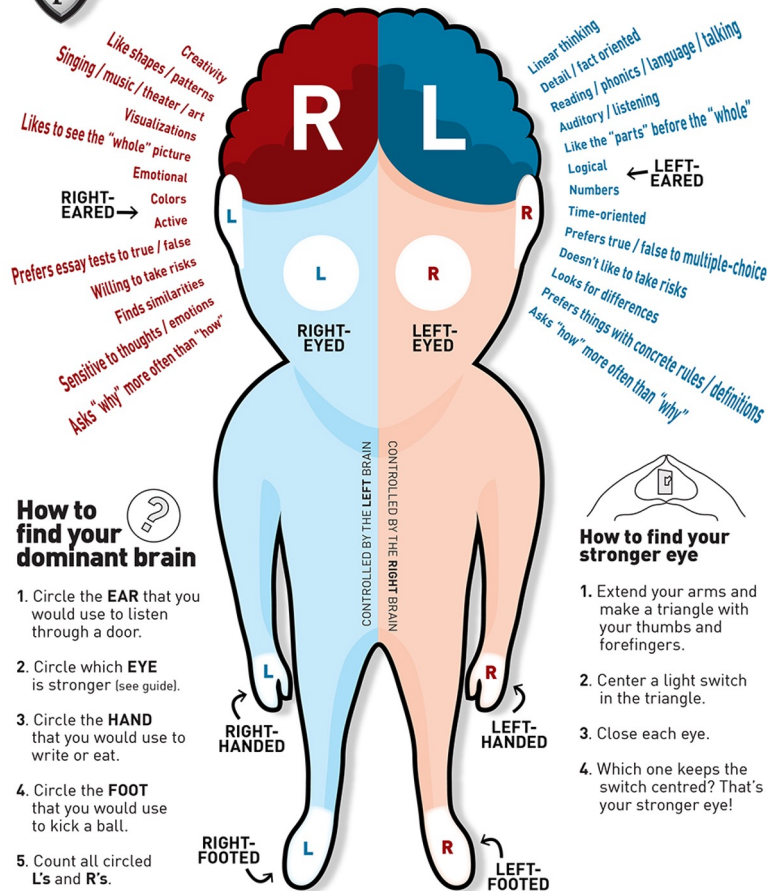
- I prefer hands-on experience and experimenting.
- I prefer reading instructions or following a clear guide.

4. In your free time, what type of activities do you enjoy most?

- Creative hobbies, such as painting, writing, or playing music.
- Analytical activities like puzzles, math, or strategy games.



Are You Right-Brained or Left-Brained?



Mostly L's = Right brained. Mostly R's = Left brained. Equal L's and R's = Balanced brain!